



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 4

Hansen's model with two types of households

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1 Environment

There are two types of households indexed by $i \in \{1, 2\}$, with population masses n_1 and n_2 such that $n_1 + n_2 = 1$. Each household of type i maximizes

$$\max \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t u_i(c_{it}, 1 - h_{it}), \quad 0 < \beta < 1, \quad (1)$$

with the specific utility function

$$u_i(c_{it}, 1 - h_{it}) = \ln c_{it} + A_i \ln(1 - h_{it}), \quad A_i > 0. \quad (2)$$

The **only** difference between the groups is the weight on leisure. Group 1 values leisure more than group 2:

$$A_1 \equiv A > B \equiv A_2. \quad (3)$$

Here c_{it} is per-capita consumption of group i , $h_{it} \in (0, 1)$ its hours worked and $1 - h_{it}$ its leisure.

Technology is the same as in the basic model. Aggregate productivity λ_t follows

$$\lambda_{t+1} = \gamma \lambda_t + \varepsilon_{t+1}, \quad 0 < \gamma < 1, \quad \mathbb{E} \varepsilon_t = 1 - \gamma, \quad (4)$$

so that $\mathbb{E} \lambda_t = 1$ and $\lambda_{t+1} > 0$.

2 The firm

Because there are two types of agents, we can no longer plug the production function directly into the utility of a single planner. We introduce a **competitive firm** that produces the single good with the Cobb–Douglas technology

$$Y_t = f(\lambda_t, K_t, H_t) = \lambda_t K_t^\theta H_t^{1-\theta}, \quad 0 < \theta < 1, \quad (5)$$

where K_t and H_t are *aggregate* capital and labor. The firm rents capital at price r_t and hires labor at wage w_t , and maximizes profits $\Pi_t = Y_t - r_t K_t - w_t H_t$. Under perfect competition, each factor is paid its marginal product:

$$r_t = \frac{\partial Y_t}{\partial K_t} = \theta \lambda_t K_t^{\theta-1} H_t^{1-\theta} = \theta \frac{Y_t}{K_t}, \quad (6)$$

$$w_t = \frac{\partial Y_t}{\partial H_t} = (1 - \theta) \lambda_t K_t^\theta H_t^{-\theta} = (1 - \theta) \frac{Y_t}{H_t}. \quad (7)$$

3 The household problem

Each household of type i owns its own capital stock k_{it} , rents it to the firm, supplies labor h_{it} , and chooses consumption and investment. Its budget constraint is

$$c_{it} + i_{it} = w_t h_{it} + r_t k_{it}, \quad (8)$$

and its capital evolves according to

$$k_{i,t+1} = (1 - \delta)k_{it} + i_{it}, \quad 0 < \delta < 1. \quad (9)$$

Combining (8)–(9), investment is determined as a residual and the budget constraint becomes

$$c_{it} + k_{i,t+1} = w_t h_{it} + (r_t + 1 - \delta) k_{it}. \quad (10)$$

3.1 Bellman equations

Unlike the basic model, here each household takes prices $\{w_t, r_t\}$ as given: it does not internalize the production function. The individual state is its capital k_{it} ; to forecast future prices the household additionally needs the aggregate state (K_t, λ_t) . The Bellman equation of type i is

$$V_i(k_{it}, K_t, \lambda_t) = \max_{c_{it}, h_{it}} \left\{ \ln c_{it} + A_i \ln(1 - h_{it}) + \beta \mathbb{E}_t [V_i(k_{i,t+1}, K_{t+1}, \lambda_{t+1}) \mid \lambda_t] \right\} \quad (11)$$

subject to the budget constraint (10), the process for λ_t (4), and the law of motion of aggregate capital K_t , which is perceived by the households in contrast to the production function. Using (10) to substitute consumption, $c_{it} = w_t h_{it} + (r_t + 1 - \delta)k_{it} - k_{i,t+1}$, the control variables are $k_{i,t+1}$ and h_{it} .

Remark. There are two Bellman equations, one per group. The equilibrium conditions are then imposed so we can work with the resulting system.

3.2 First-order conditions

Differentiating (11) with respect to $k_{i,t+1}$ and h_{it} , and applying the envelope theorem (Benveniste–Scheinkman) $\partial V_i / \partial k_{it} = (r_t + 1 - \delta) / c_{it}$, we obtain, for each $i \in \{1, 2\}$:

- **Euler equation** (intertemporal consumption/saving choice):

$$\frac{1}{c_{it}} = \beta \mathbb{E}_t \left[\frac{r_{t+1} + (1 - \delta)}{c_{i,t+1}} \mid \lambda_t \right]. \quad (12)$$

- **Intratemoral leisure–consumption condition** (labor/leisure choice):

$$\frac{A_i}{1 - h_{it}} = \frac{w_t}{c_{it}} \iff A_i c_{it} = w_t (1 - h_{it}). \quad (13)$$

Condition (13) is where heterogeneity becomes relevant: for the same wage w_t and consumption, the group with the higher preference for leisure (A) chooses to work less.

4 Aggregation

Aggregate variables are the population-weighted sums of each group's choices:

$$K_t = n_1 k_{1t} + n_2 k_{2t}, \quad (14)$$

$$H_t = n_1 h_{1t} + n_2 h_{2t}, \quad (15)$$

$$C_t = n_1 c_{1t} + n_2 c_{2t}, \quad (16)$$

$$I_t = n_1 i_{1t} + n_2 i_{2t}. \quad (17)$$

In particular, aggregate capital (14) —which enters the production function (5) and the factor prices (6)–(7)— are the sum of what each group provides.

5 Equilibrium conditions

A recursive competitive equilibrium consists of plans $\{c_{it}, h_{it}, k_{i,t+1}\}_{i=1,2}$, aggregates $\{Y_t, K_t, H_t, C_t, I_t\}$ and prices $\{r_t, w_t\}$ such that:

- (i) given prices, each group's plan solves its own Bellman eq. so it satisfies the FOCs (12)–(13) and the budget constraint (10);
- (ii) the firm maximizes profits (6)–(7)
- (iii) the factor markets clear through the aggregations (14)–(15)
- (iv) and the economy's resource constraint holds.

Summing the individual budget constraints (8) weighted by n_i , using the factor payments (6)–(7) (which imply $r_t K_t + w_t H_t = Y_t$) and the aggregations, we obtain the **aggregate resource constraint**

$$Y_t = C_t + I_t, \quad (18)$$

and the corresponding law of motion of **aggregate capital**

$$K_{t+1} = (1 - \delta)K_t + I_t. \quad (19)$$

Remark. (19) is identical in form to that of the basic model, but the dynamics now operate on aggregate capital (14), the sum of the two groups' holdings.

5.1 Summary of the system

The equilibrium is described by the following system (for $i = 1, 2$):

$$Y_t = \lambda_t K_t^\theta H_t^{1-\theta} \quad (\text{production}) \quad (20)$$

$$r_t = \theta Y_t / K_t, \quad w_t = (1 - \theta) Y_t / H_t \quad (\text{factor prices}) \quad (21)$$

$$\frac{1}{c_{it}} = \beta \mathbb{E}_t \left[\frac{r_{t+1} + (1 - \delta)}{c_{i,t+1}} \mid \lambda_t \right] \quad (\text{Euler, 2 eq.}) \quad (22)$$

$$A_i c_{it} = w_t (1 - h_{it}) \quad (\text{leisure-consumption, 2 eq.}) \quad (23)$$

$$c_{it} + k_{i,t+1} = w_t h_{it} + (r_t + 1 - \delta) k_{it} \quad (\text{budget, 2 eq.}) \quad (24)$$

$$K_t = n_1 k_{1t} + n_2 k_{2t}, \quad H_t = n_1 h_{1t} + n_2 h_{2t} \quad (\text{aggregation}) \quad (25)$$

$$Y_t = C_t + I_t, \quad C_t = n_1 c_{1t} + n_2 c_{2t} \quad (\text{resources}) \quad (26)$$

$$\lambda_{t+1} = \gamma \lambda_t + \varepsilon_{t+1} \quad (\text{stochastic process}) \quad (27)$$

6 Steady state

In the steady state all variables are constant and there are no productivity shocks, so we can assume $\bar{\lambda} = 1$. The conditions reduce to:

$$\frac{1}{\beta} = \bar{r} + (1 - \delta), \quad (28)$$

$$\bar{r} = \theta \bar{K}^{\theta-1} \bar{H}^{1-\theta}, \quad \bar{w} = (1 - \theta) \bar{K}^\theta \bar{H}^{-\theta}, \quad (29)$$

$$A_i \bar{c}_i = \bar{w} (1 - \bar{h}_i), \quad i = 1, 2, \quad (30)$$

$$\bar{c}_i = \bar{w} \bar{h}_i + (\bar{r} - \delta) \bar{k}_i, \quad i = 1, 2, \quad (31)$$

$$\bar{K} = n_1 \bar{k}_1 + n_2 \bar{k}_2, \quad \bar{H} = n_1 \bar{h}_1 + n_2 \bar{h}_2. \quad (32)$$

The Euler equation (28) is the same for both groups since it does not depend on A_i , and pins down the return on capital,

$$\bar{r} = \frac{1}{\beta} - (1 - \delta), \quad (33)$$

which in turn pins down the aggregate capital-labor ratio $\bar{K}/\bar{H}$ through (29). The wage $\bar{w}$ is then determined. Given $\bar{r}$ and $\bar{w}$, conditions (30)–(31) characterize each group's allocation.

7 Wealth comparison across groups

In this model a group's wealth is the capital it owns, $\bar{k}_i$. From the steady-state budget constraint (31), where $\bar{i}_i = \delta \bar{k}_i$, we solve directly

$$\boxed{\bar{k}_i = \frac{\bar{c}_i - \bar{w} \bar{h}_i}{\bar{r} - \delta}, \quad \bar{r} - \delta = \frac{1}{\beta} - 1 > 0.} \quad (34)$$

Comparing the two groups,

$$\bar{k}_1 - \bar{k}_2 = \frac{(\bar{c}_1 - \bar{c}_2) - \bar{w}(\bar{h}_1 - \bar{h}_2)}{\frac{1}{\beta} - 1} < 0 \quad (35)$$

Interpretation. Equation (34) says that a group's capital is proportional to the part of its consumption that is *not* financed by labor income: the group that consumes more than it earns from labor ($\bar{c}_i > \bar{w}\bar{h}_i$) sustains that gap with capital income and therefore holds more capital. Since group 1 values leisure more ($A > B$), from (30) it works less ($\bar{h}_1 < \bar{h}_2$) and receives less labor income, which tends to leave it with lower capital wealth.

Note on the distribution. With a common discount factor β and logarithmic utility, the two Euler equations collapse in the steady state to the same condition (28): they do not discriminate between groups. As a result, the system pins down the aggregates (in particular $\bar{K}/\bar{H}$) but leaves the split of capital between the two groups undetermined: the wealth distribution depends on initial conditions, i.e. the initial endowments k_{10}, k_{20} . Only with these endowments, wealth can be compared along that distribution. However, equation (35) remains valid as an equilibrium relationship.